

HAOTIAN YIN

eileenyinhtee@gmail.com
www.linkedin.com/in/eileenyinht

SUMMARY

Ph.D. candidate in Computer Science at NJIT, focusing on 3D reconstruction, generative AI, and neural surface representation. My research develops implicit neural networks for 3D shape reconstruction and representation, achieving high-fidelity geometry and enabling applications in inverse problems and 3D generation. Experienced with deep learning frameworks (PyTorch), 3D data processing and evaluation, and publishing at top conferences such as SIGGRAPH Asia.

EDUCATION

New Jersey Institute of Technology (NJIT), College of Computing	New Jersey, U.S.
PhD in Computer Science, GPA 3.96/4.0, Advisor: <i>Przemyslaw Musialski</i>	09/2020-Present
Relevant Coursework: Machine learning, Data Mining, Data Analysis with R, NLP, optimization.	
Jilin University (Double Major)	Changchun, China
College of Mathematics: Bachelor in Information and Computational Science	09/2016-07/2020
College of Computer Science: Bachelor in Computer Application	09/2017-07/2020

HIGHLIGHTED RESEARCH EXPERIENCE

PhD. Student | Graphics and Neural Surface Representation NJIT, 09/2021-Present

[SIGGRAPH ASIA'25] A Finite Difference Approximation of Second Order Regularization for Neural-SDFs.

- Developed a lightweight optimization framework replacing computationally expensive second-order derivatives with finite-difference operators, improving model training efficiency and numerical stability.
- Enabled large-scale neural modeling using only first-order gradients — an approach generalizable to AIGC and reinforcement learning systems requiring interpretable and efficient optimization.

[Pacific Graphics'25] FlatCAD: Fast Curvature Regularization of Neural SDFs for CAD Models

- Proposed an off-diagonal loss design that stabilizes model training and improves convergence speed by 2×.
- Demonstrated the framework's ability to transfer across domains — from 3D surface learning to general deep network regularization and AI model optimization under limited compute budgets.

[ISVC'25] Scheduling Curvature-Aware Loss with Annealed Weights

- Designed an adaptive scheduling strategy that dynamically adjusts regularization weights during training to maintain convergence stability. Improved overall model accuracy by 35%, offering a general mechanism for loss balancing in large-scale AI and multi-agent learning scenarios.

Undergrade Student | Statistics, Modelling and Machine Learning Jilin University, 09/2016- 07/2020

Collection of Synthetic RNA-based Thermosensors with Different Sensing Temperature

-2018 International Genetically Engineered Machine competition (iGEM); **Gold Award Winner**

Leader in modeling, Team Jilin_China, http://2018.igem.org/Team:Jilin_China

- Built statistic models describing switch behavior of RNA-based thermosensors and fit logistic curves.
- Built an intelligent screening system to screen desirable RNA-based thermosensors, and predicted the validity of thermosensors with different nucleotide sequences by using a random forest algorithm.

PROFESSIONAL EXPERIENCE

Taiping Insurance Corporation Shenzhen, China, 07/2020-09/2020

Risk Actuary Intern | Group of Analysis and Monitoring

- Conducted large-scale insurance data analysis using statistical modeling and data mining to identify key predictors of claim risk through correlation and regression (ELES).

- Built and optimized predictive models for loss forecasting and policy evaluation, delivering actionable insights for risk-control decisions.

Sinovation Ventures & Biomind

Beijing, China, 07/2019-08/2019

Algorithm Developer | DeeCamp Artificial Intelligence Program | Python, Pytorch, Computer Vision

- Built AI-driven medical image segmentation pipeline (U-Net, YOLOv3)
- Optimized data pre-processing and augmentation for model generalization in large-scale medical datasets, improving accuracy from 69 % $\rightarrow$ 93 %.

Teaching Assistant in NJIT

NJIT, 09/2021-Present

Courses: Machine Learning, AI for Robotics, Cloud Computing, Python, MATLAB

PUBLICATIONS

- **H. Yin**, A. Plocharski, M. J. Wlodarczyk, M. Kida and P. Musialski, “A Finite Difference Approximation of Second Order Regularization of Neural-SDFs.” **SIGGRAPH Asia** Technical Communications, 2025.
- **H. Yin**, A. Plocharski, M. J. Wlodarczyk, M. Kida and P. Musialski, “FlatCAD: Fast Curvature Regularization of Neural SDFs for CAD Models.” **Computer Graphics Forum (Proc. Pacific Graphics)**, 2025.
- **H. Yin** and P. Musialski, “Scheduling the Off-Diagonal Weingarten Loss for CAD-Type Neural SDF Reconstruction.” International Symposium on Visual Computing (ISVC), 2025.
- **H. Yin** and P. Musialski, “Shrinking: Reconstruction of parameterized surfaces from signed distance fields.” IEEE International Conference on Machine Learning and Applications (ICMLA), 2024.
- **H. Yin**, Z. Li, H. Hu, L. Tian and Y. Huang, Model and Measurement of China's Urban and Rural Minimum Living Security Standards. *Electronic Technology & Software Engineering*, ISSN 2095-5650, Sep 2019
- **H. Yin**, H. Hu, L. Tian and Z. Li, Ocean Wave Propagation Model Based on Ocean Wave Correction. *Digital Communication World*, ISSN 1672-7274, May 2018

SKILLS

Research Areas: Machine Learning, Deep Learning, Optimization, Generative Models(AIGC), Computer Graphic

Programming Languages: Python (Pytorch), R, SQL, C++, MATLAB, Pascal, C

Data & Mathematics: Statistical Modeling, Probability & Statistics, Differential Geometry, Finite Difference Methods

Tools: Model Deployment, Cloud Computing (AWS, GCP), Git, Blender, SPSS

Language: English (Fluent), Chinese (Native), Cantonese (Basic), Japanese (Basic)

AHIEVEMENTS & COLLABORATION

Modeling Contest Winner: Meritorious Winner in 2019 Mathematical Contest in Modeling — recognized for analytic modeling and optimization under constraints.

Volunteer Service: Served as student volunteer at SIGGRAPH 2023, I3D 2025 and session chair at ICMLA 2024

Cross-Cultural Collaboration: Participated in research teams across Europe, U.S., and Asia; proficient in coordinating across time zones and adapting to diverse research cultures. Collaboration with Warsaw University of Technology on Neural Shape Processing.

Self-driven and Adaptable: with a strong interest in emerging technologies and motivation to turn research ideas into real-world 3D vision applications.